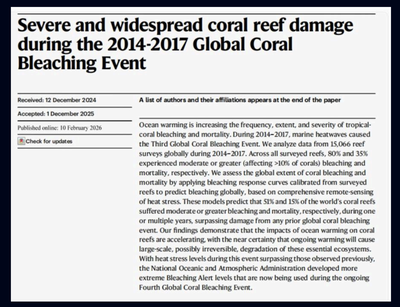

Bleach Alert
A Cooperative Multiplayer Game Dedicated To Addressing Coral Bleaching.

INSPIRATION





Severe and widespread coral reef damage during the 2014-2017 Global Coral Bleaching Event

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Ocean warming is increasing the frequency, extent, and severity of tropical-coral bleaching and mortality. During 2014–2017, marine heatwaves caused the Third Global Coral Bleaching Event. We analyze data from 15,066 reef surveys globally during 2014–2017. Across all surveyed reefs, 80% and 35% experienced moderate or greater (affecting >10% of corals) bleaching and mortality, respectively. We assess the global extent of coral bleaching and mortality by applying bleaching response curves calibrated from surveyed reefs to predict bleaching globally, based on comprehensive remote-sensing of heat stress. These models predict that 51% and 15% of the world's coral reefs suffered moderate or greater bleaching and mortality, respectively, during one or multiple years, surpassing damage from any prior global coral bleaching event. Our findings demonstrate that the impacts of ocean warming on coral reefs are accelerating, with the near certainty that ongoing warming will cause large-scale, possibly irreversible, degradation of these essential ecosystems. With heat stress levels during this event surpassing those observed previously, the National Oceanic and Atmospheric Administration developed more extreme Bleaching Alert levels that are now being used during the ongoing Fourth Global Coral Bleaching Event.



AFTER — CORAL BLEACHING

What Happens When The Colors Of The Ocean Begin To Disappear?

BACKGROUND

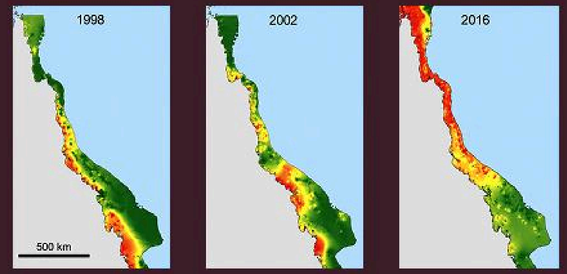
In Recent Years, Rising Global Ocean Temperatures Have Caused Coral Bleaching Events To Spread At An Unprecedented Rate.





Once Vibrant And Full Of Life, Coral Reefs Are Gradually Losing Their Color, Ecological Functions, And Ability To Survive.

Global Situation



Between 1998 And 2016, The Extent Of Coral Bleaching Expanded Significantly And Its Severity Intensified; By 2016, The Phenomenon Had Evolved From Localized, Sporadic Occurrences Into **Widespread, High-Intensity Bleaching.**

The Core Call Of The United Nations Sustainable Development Goal (SDG) 14 Is To Conserve And Sustainably Use The Oceans, Seas And Marine Resources To Support The Well-Being Of Current And Future Generations.



The Impact Of Coral Bleaching On Organisms

At The Aldabra Atoll In Seychelles, Despite Minimal Local Human Impact, **50%** Of The Coral Reefs Have Died Due To Bleaching.

Extensive Fish Surveys Conducted In 2015 And 2020 Revealed A **55–60%** Decline In Overall Abundance Across Both Shallow And Deep Reefs, A 69% Reduction In Biomass On Shallow Reefs, And A Decrease In Biomass Across All Functional Groups.



Brainstorm

a cooperative game about coral bleaching protection

·SITUATION :
GLOBAL CORAL BLEACHING CRISIS

Rising ocean temperatures and environmental stress are causing coral reefs around the world to bleach and decline.



·INSIGHT: THE AWARENESS GAP

Most visitors only see the beauty of reefs, but lack a deep understanding of the crisis and the effort behind restoration.

SEE	HEAR	UNDERSTAND	TAKE ACTION
The Reef	The Problem	The System	Protect the Reef

DESIGN OPPORTUNITY

How might we turn coral bleaching from something visitors simply "hear about" into something they can actively **experience and understand**?

·WHY A GAME?

Compared to traditional education, games create immersive, interactive experiences that engage players emotionally and cognitively.

PASSIVE EDUCATION	INTERACTIVE EXPERIENCE
Watch · Read · Forget	Explore · Decide · Cooperate · Restore · Remember

CORAL BLEACHING BY THE NUMBERS

1.1°C	The ocean has warmed by about 1.1°C since pre-industrial times.
90%	Up to 90% of coral reefs could disappear by 2050.
50%+	More than 50% of the world's reefs have experienced bleaching in the past 30 years.
25%	Coral reefs support 25% of all marine life.

·SOME SOLUTIONS

AUSTRALIA
Great Barrier Reef Marine Park
Improved water quality, crown-of-thorns starfish control, coral nurseries.

INDONESIA
Coral Triangle Initiative
Cross-border cooperation, sustainable fishing, coral restoration.

UNITED STATES
NOAA Coral Reef Conservation Program
Monitoring, research, and community restoration projects.

MALDIVES
Coral Restoration & Protection
Coral nurseries, temperature monitoring, anchor damage prevention.

INTERNATIONAL EFFORTS
UN Decade on Ecosystem Restoration
Global collaboration, funding, and knowledge sharing to restore our ocean.

GAME REFERENCE

Subnautica
— Environmental & Equipment Design

Design Reference

Subnautica inspired the **spatial and structural design** of the underwater environment. Its submarines, underwater bases, diving equipment, and functional devices provided references for building a believable marine workspace.







Key Takeaways

- Underwater Vessel Structure
- Functional Equipment
- Diving Facilities
- Marine Environment

Barotrauma
— Cooperative Gameplay



Mechanism Reference

Barotrauma inspired the game's **role-based cooperative system**. Players take on different responsibilities, operate specialized equipment, share information, and respond together to unexpected situations.





Key Takeaways

·Role Division
· Team Cooperation
· Task Dependency
· Emergency Response





DREDGE

Key Takeaways

·UI Design
·Visual Motion Effects

Character Setting And Coordination



DIVER

Explore, collect data and transport reef modules safely.



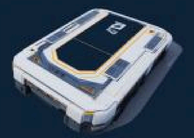
HATCHERY TECH

Cultivate corals, analyze data and prepare reef modules.



CAPTAIN

Navigate the vessel, protect the crew and coordinate logistics.



Workbench

Provide various forms of support

1.MISSION PREP
Plan & Equip

- Review Mission Objectives&Site Info
- Check & Prepare Diving Equipment
- Prepare Coral Fragments & Tools
- Check Data Tools & 3D Printer Materials
- Review Weather & Sea Conditions
- Plan Route & Mission Logistics
- Brief Crew & Assign Roles

2.FIELD OPERATION
Explore & Collect

- Dive & Explore Reef Site
- Collect Ecological Data & Images
- Identify Suitable Restoration Areas
- Receive Samples & Environmental Data
- Analyze Data Onboard
- Select & Prepare Coral Fragments
- Navigate To Operation Site
- Maintain Vessel Position & Safety
- Communicate & Coordinate Team

3.REEF RESTORATION
Build & Restore

- Transport Artificial Reef Modules
- Deploy & Install Reef Structures
- Assist With Coral Planting
- Attach & Secure Coral Fragments
- 3D Print Reef Modules If Needed
- Tag & Record Coral Transplantation
- Support Deployment Operations
- Ensure Crew Safety During Operations
- Adjust Position As Needed

4.MONITOR & IMPROVE
Evaluate & Adapt

- Post-Dive Inspection & Site Monitoring
- Report Observations & Issues
- Recommend Next Steps
- Analyze Growth & Environmental Data
- Evaluate Restoration Success
- Suggest Improvements & Next Plan
- Review Mission Outcomes
- Adjust Routes & Future Plans
- Log Results & Mission Complete

TEAMWORK LOOP — Continuous communication • Share updates • Support each other • Adapt to changes • Protect the reef

EMERGENCY EVENT
React&Resolve

POSSIBLE EVENTS
- Sudden Weather Change
- Strong Currents
- Equipment Failure
- Diver Injury
- Vessel Malfunction

EMERGENCY RESPONSE
(Team Collaboration)
- Assess The Situation
- Communicate Quickly
- Execute Safety Protocol

ROLE ACTIONS

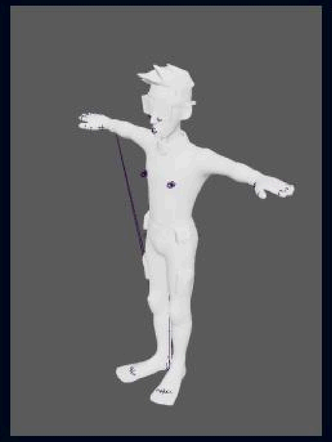
DIVER (Blue)
- Signal & Report
- Return To Vessel
- Ensure Personal Safety

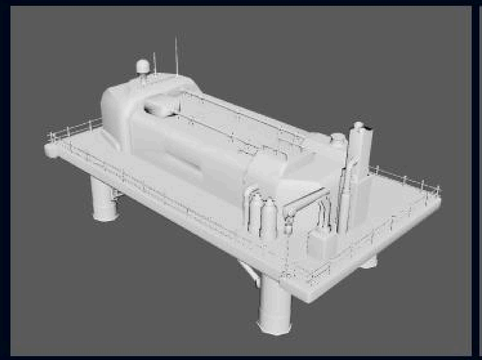
HATCHERY TECH (Green)
- Secure Samples & Equipment
- Support First Aid
- Backup Data

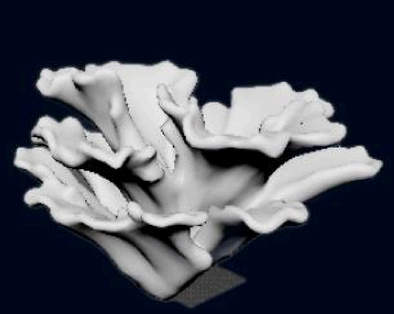
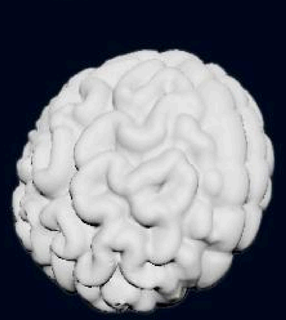
CAPTAIN (Orange)
- Stabilize Vessel
- Make Safety Decisions
- Contact For Support If Needed

RESUME MISSION or ABORT MISSION

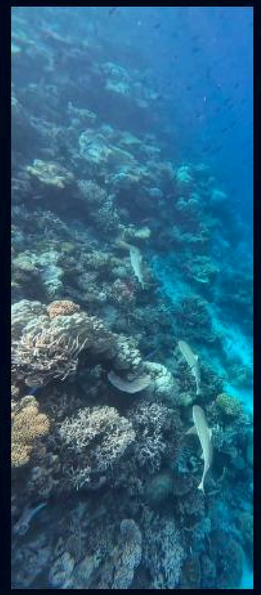
Development Process

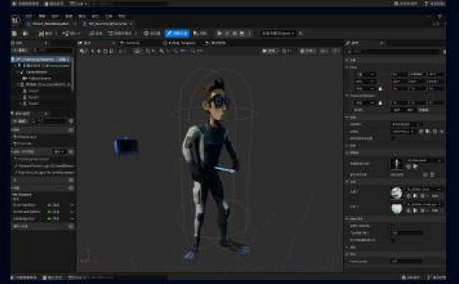



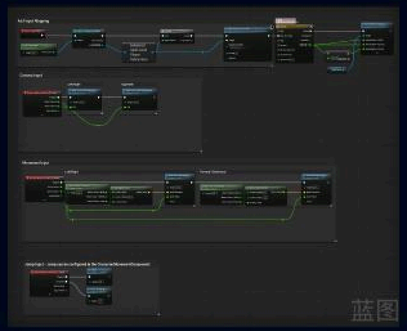




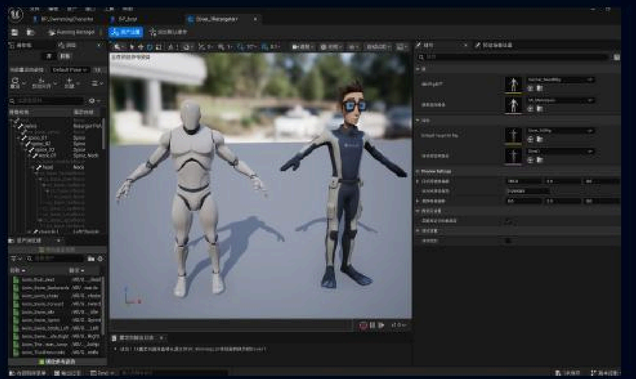







I created the main gameplay systems in Unreal Engine using Blueprints, including player interaction, coral cultivation, and planting mechanics. I used Foliage Mode to efficiently build the underwater environment and distribute corals, rocks, and marine vegetation across the seabed. The main environmental assets and characters were modelled separately and then imported into Unreal Engine, where I assembled the final scenes, adjusted materials, lighting, and atmosphere, and integrated the characters with the gameplay systems.






Gameplay





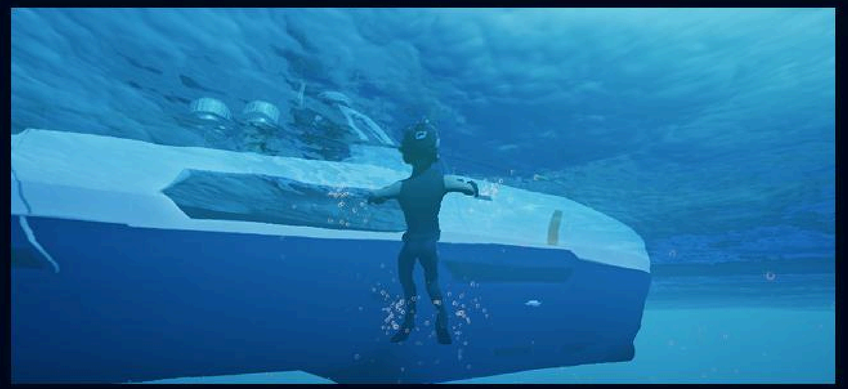












After completing the coral cultivation and planting system, I plan to expand the project with additional gameplay focused on **marine conservation**. Future features may include monitoring coral health, cleaning polluted areas, restoring damaged habitats, and protecting marine life from environmental threats. These activities will gradually show how different conservation efforts contribute to the recovery of the underwater ecosystem.